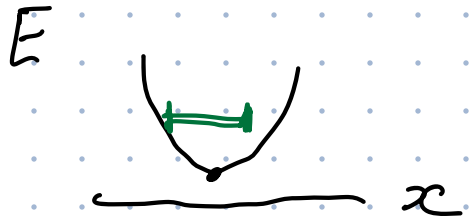
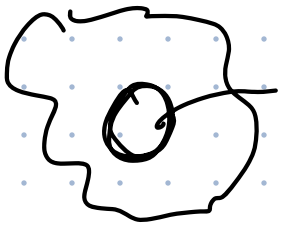
ENERGY

$E(x)$

↑ CONFIGURATION





NUCLEUS

$$F = -\frac{\partial E}{\partial x} \quad \text{OR} \quad \vec{F} = -\nabla E$$

$\Rightarrow$ 

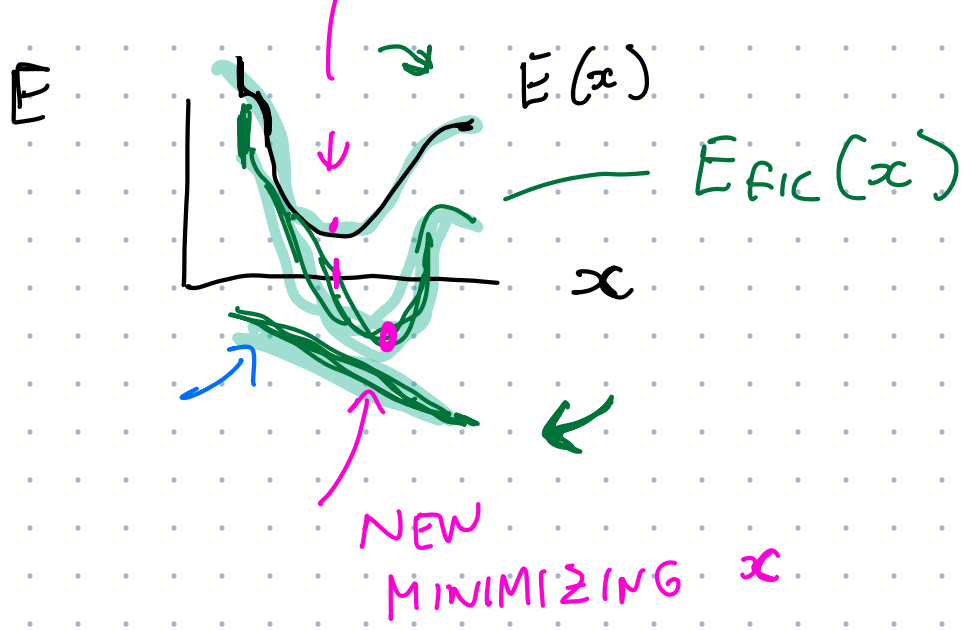
x WILL MINIMIZE $E(x)$



$$E_{fic}(x) = E(x) - x \cdot F_{EXT}$$

FICTITIOUS POTENTIAL

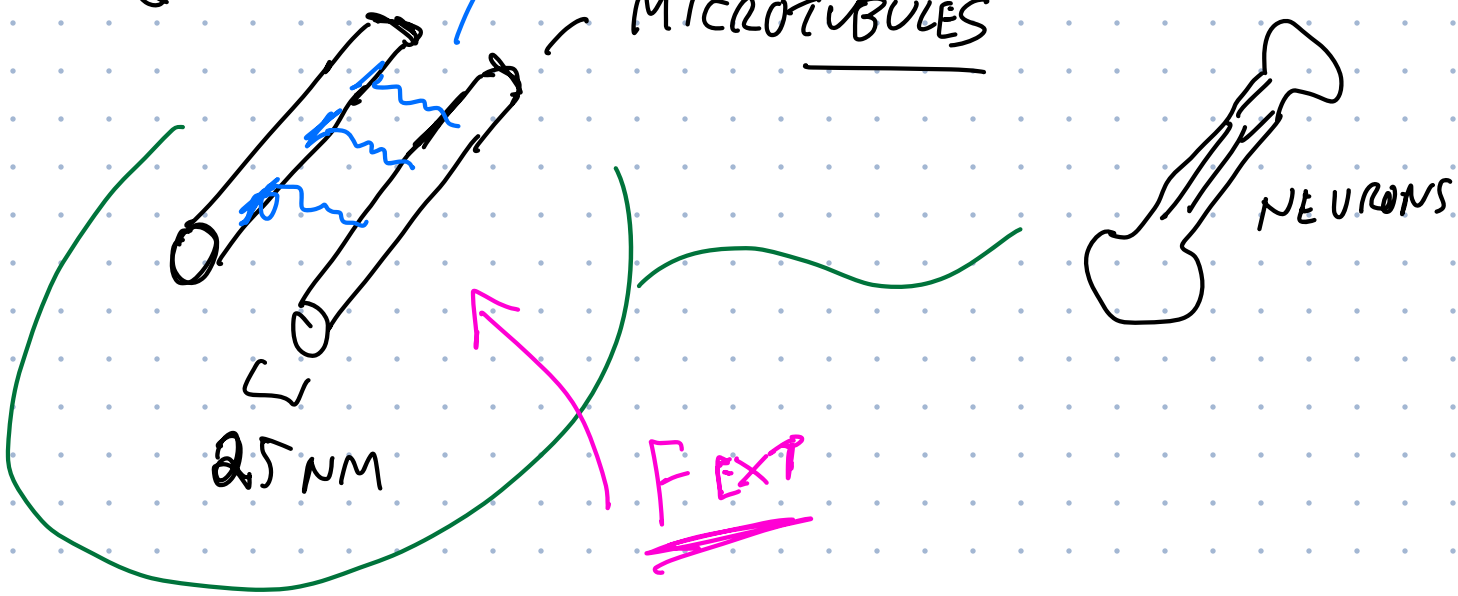
MINIMIZING x

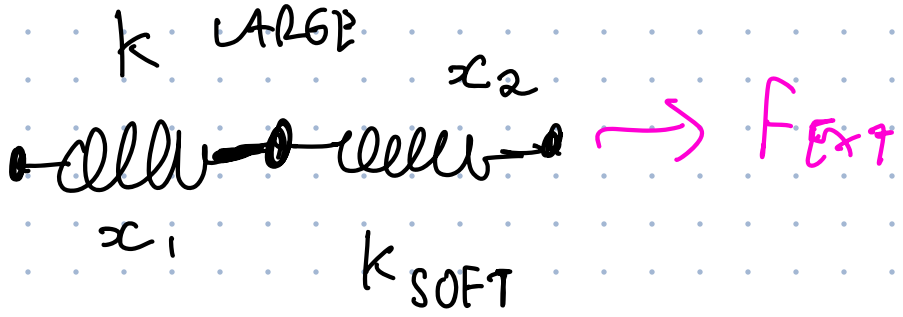


PS 1

TAU → UCI SWARUP GROSS

MICROTUBULES





$E(x_1, x_2)$

$E(x_1, x_2)$

$E = ?$ $\frac{1}{2} k_1 x_1^2 + \frac{1}{2} k_2 x_2^2 - F \Delta x$

Δx

$\Delta x =$

$?$ $(\bar{x}_1 - x_1)^2$ $(\bar{x}_2 - x_2)^2$ $\left((\bar{x}_1 - x_1) + (\bar{x}_2 - x_2)\right)$

$E = \frac{1}{2} k_1 (\bar{x}_1 - x_1)^2 + \frac{1}{2} k_2 \left(\bar{x}_2 - (x_1 - \bar{x}_2 - \Delta x)\right)^2$

POS ?

$- F_{Ext} (\Delta x_1 + \Delta x_2)$